Aim: Implement a Syntax Analyzer: Write a Context-Free Grammar (CFG) for the defined programming language constructs. Use Lex and YACC tools to validate the syntax of the input program.

Code :

Calc.l File :

```
%{
#include <stdlib.h>
#include "calc.tab.h"
%}

%option noyywrap

%%
[0-9]+      { yylval = atoi(yytext); return NUM; }
[ \t]+      { /* skip spaces */ }
\n          { return EOL; }
.           { return yytext[0]; }
%%
```

Calc.y File :

```
%{
#include <stdio.h> int
yylex(void);
void yyerror(const char *msg);
%}

%token NUM EOL

%left '+' '-'
%left '*' '/'

%%
program:
      /* empty */
    | program line
    ;
```

```
line:
      EOL
    | expr EOL      { printf("Answer: %d\n", $1); }
    | error EOL     { yyerrok; }
    ;

expr:
      expr '+' expr { $$ = $1 + $3; }
| expr '-' expr { $$ = $1 - $3; }
    | expr '*' expr { $$ = $1 * $3; }
    | expr '/' expr {
                      if ($3 == 0) { printf("Division by zero\n"); $$ = 0; }
else $$ = $1 / $3;
                  }
    | '(' expr ')'  { $$ = $2; }
    | NUM           { $$ = $1; }
    ;
%%

void yyerror(const char *msg)
{
    printf("Syntax error: invalid expression\n");
}

int main(void)
{
    printf("Enter expressions (Ctrl+D to quit):\n");
yyparse();     return 0;
}
```

Output :

```
(base) shivampatel@Shivams-MacBook-Pro p4 % bison -d calc.y
flex calc.l
gcc calc.tab.c lex.yy.c -o calc
./calc
Enter expressions (Ctrl+D to quit):
(4+5)*2
Answer: 18
2+*3
Syntax error: invalid expression
(base) shivampatel@Shivams-MacBook-Pro p4 %
```